Relatorio de Treinamento de Modelo de Classificacao

Classificacao de plantas de milho: healthy / unhealthy

Arquitetura: YOLO26n-cls (Ultralytics) — fine-tuning a partir de pesos pre-treinados em ImageNet
Dataset exportado do Roboflow, formato Folder Structure

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1. Dataset e Estratificacao

Split	Healthy	Unhealthy	Total	% do dataset	% unhealthy no split
Train	15905	1744	17649	70.0%	9.9%
Valid	4553	521	5054	20.0%	10.3%
Test	2267	250	2517	10.0%	9.9%
Total	**22725**	**2515**	**25240**	**100.0%**	**9.97%**

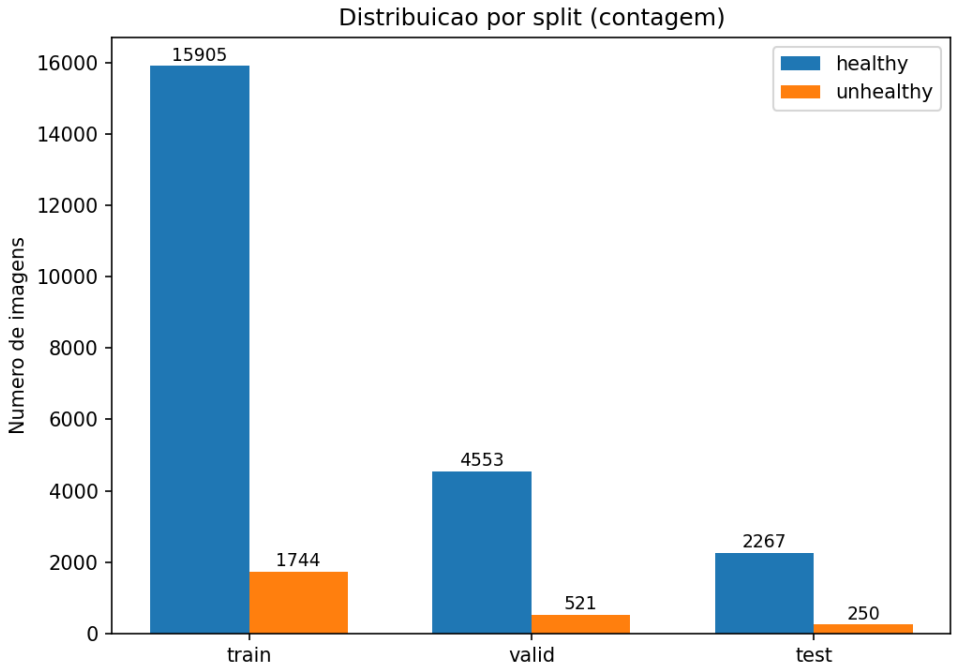


Figura 1.1 - Numero de imagens por split e classe.

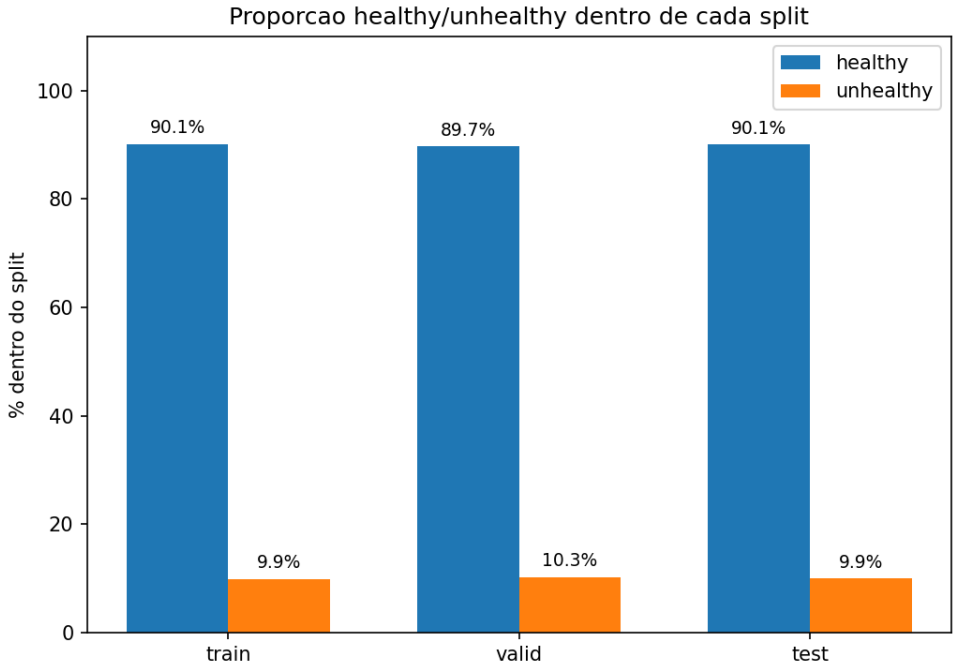


Figura 1.2 - Proporcao healthy/unhealthy dentro de cada split.

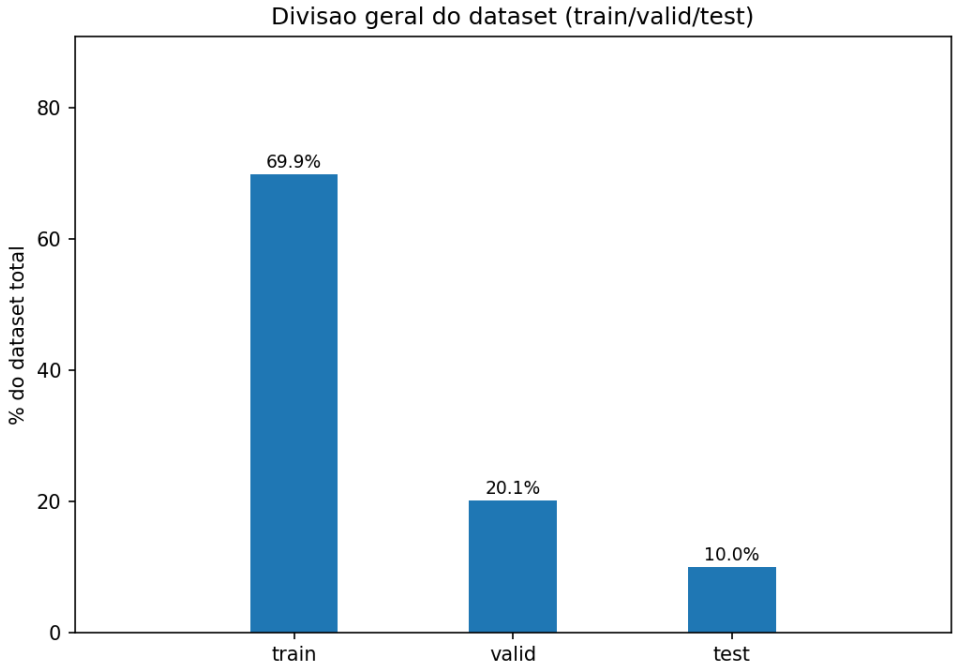


Figura 1.3 - Divisao percentual geral do dataset entre train/valid/test.

2. Configuracao do Treinamento

Parametro	Valor
Modelo	yolo26n-cls.pt (pre-treinado ImageNet)
Epocas configuradas (max)	100
Epocas efetivamente treinadas	72 (early stopping, patience=20)
Image size	224 x 224
Batch size	64
Optimizer	auto (MuSGD, selecionado automaticamente)
Warmup epochs	3
Device	GPU Tesla T4 (Google Colab)
Seed	0 (deterministic=True)

3. Curvas de Treinamento

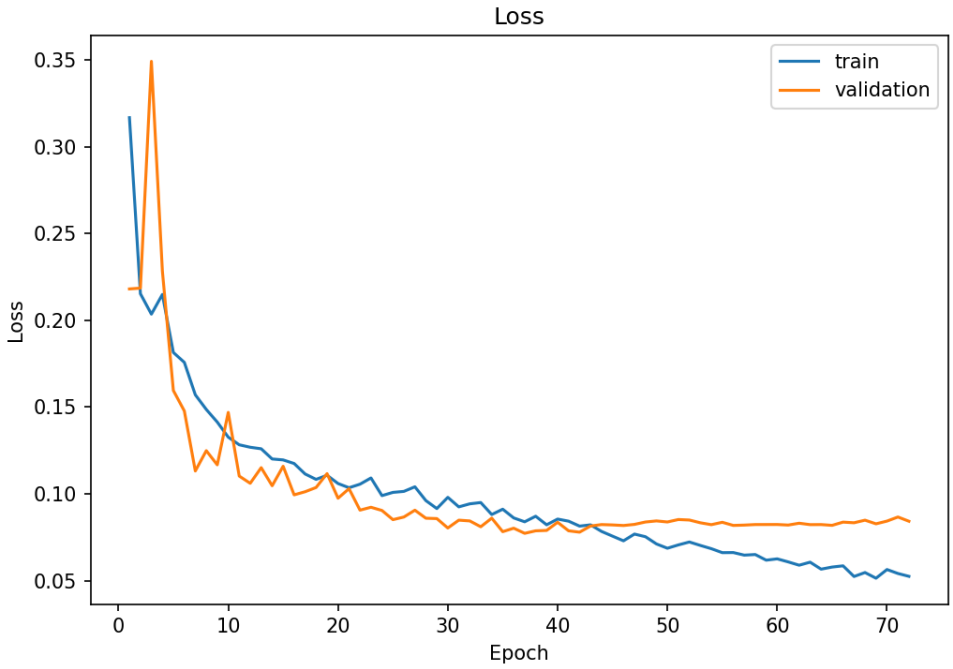


Figura 3.1 - Loss por epoca (treino vs validacao).

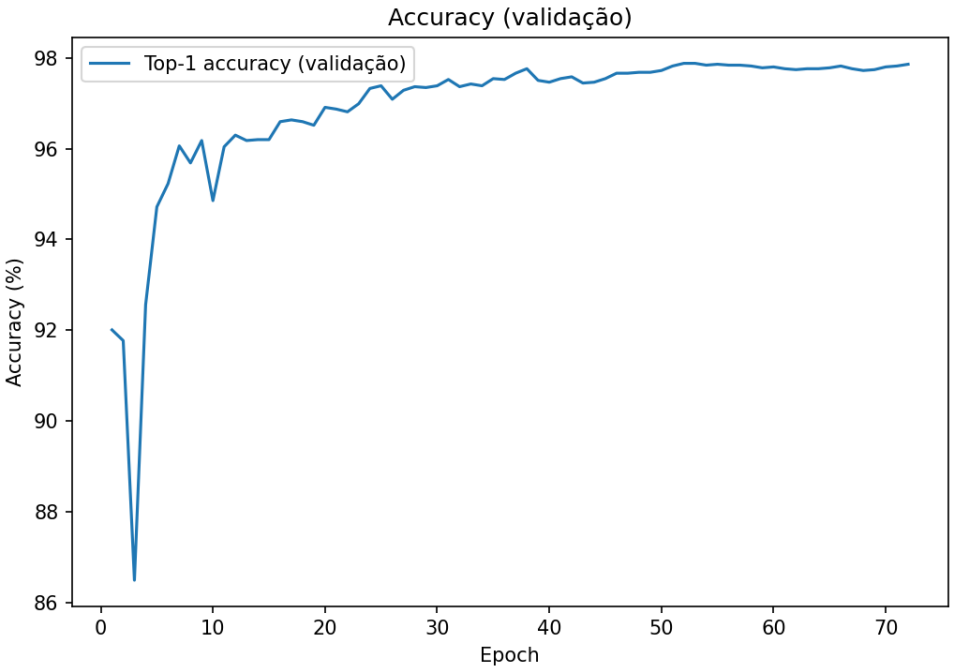


Figura 3.2 - Top-1 accuracy por epoca (split de validacao).

4. Metricas Finais por Split

Tabela 4.1 - Accuracy final por split.

Split	Accuracy (top-1)
Train	0.9929
Valid	0.9786
Test	0.9746

Tabela 4.2 - Precision, Recall e F1-score por classe — split TRAIN (17649 imagens).

Classe	Precision	Recall	F1-score	Support
healthy	0.9947	0.9974	0.9960	15905
unhealthy	0.9753	0.9518	0.9634	1744
accuracy			0.9929	17649
macro avg	0.9850	0.9746	0.9797	17649
weighted avg	0.9928	0.9929	0.9928	17649

	Previsto: healthy	Previsto: unhealthy
Real: healthy	15863	42
Real: unhealthy	84	1660

Tabela 4.3 - Precision, Recall e F1-score por classe — split VALID (5054 imagens).

Classe	Precision	Recall	F1-score	Support
healthy	0.9847	0.9916	0.9881	4533
unhealthy	0.9223	0.8656	0.8931	521
accuracy			0.9786	5054
macro avg	0.9535	0.9286	0.9406	5054
weighted avg	0.9782	0.9786	0.9783	5054

	Previsto: healthy	Previsto: unhealthy
Real: healthy	4495	38
Real: unhealthy	70	451

Tabela 4.4 - Precision, Recall e F1-score por classe — split TEST (2517 imagens).

Classe	Precision	Recall	F1-score	Support
healthy	0.9829	0.9890	0.9859	2267
unhealthy	0.8941	0.8440	0.8683	250
accuracy			0.9746	2517
macro avg	0.9385	0.9165	0.9271	2517
weighted avg	0.9741	0.9746	0.9742	2517

5. Matriz de Confusao (split test)

Tabela 5.1 - Matriz de confusao em numero de imagens.

	Previsto: healthy	Previsto: unhealthy
Real: healthy	2242	25
Real: unhealthy	39	211

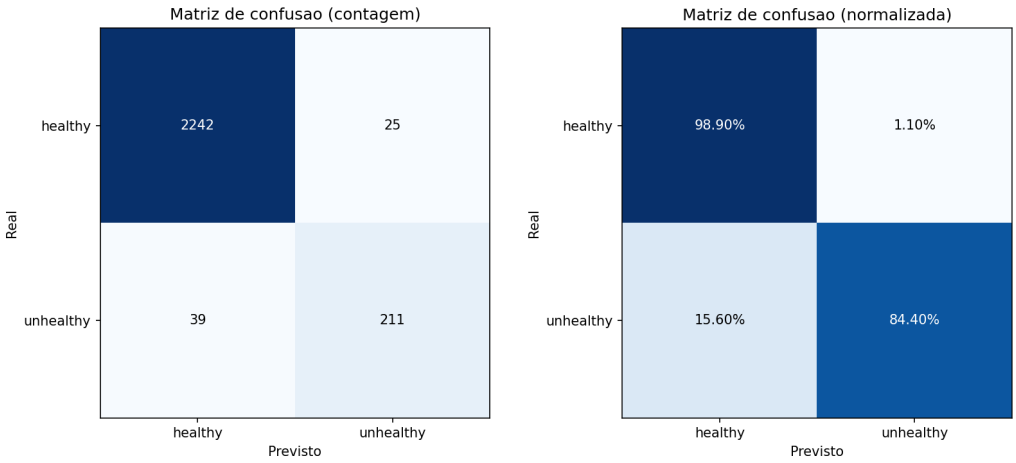


Figura 5.1 - Matriz de confusao: contagem (esquerda) e normalizada (direita).

6. Curvas ROC e Precision-Recall (classe unhealthy, split test)

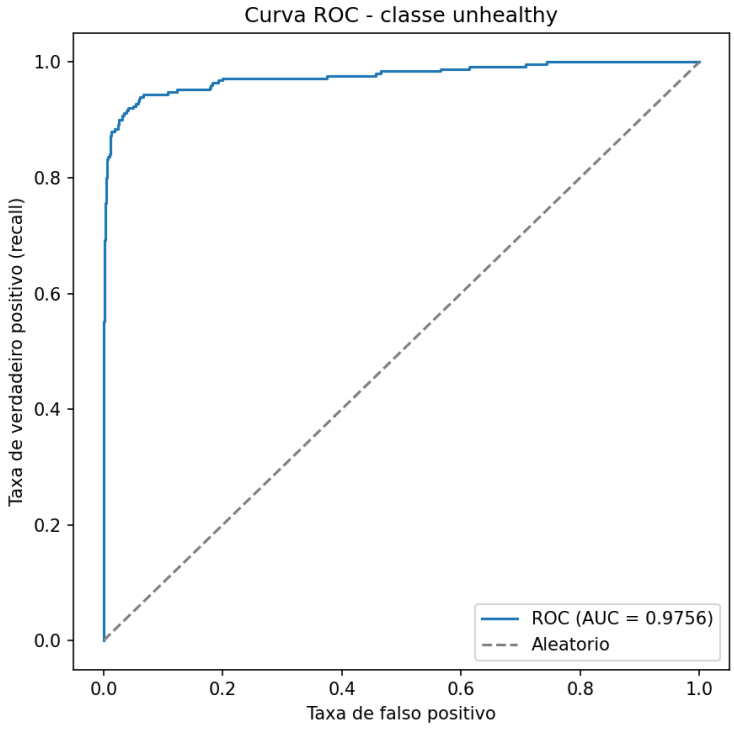


Figura 6.1 - Curva ROC. AUC = 0.9756.

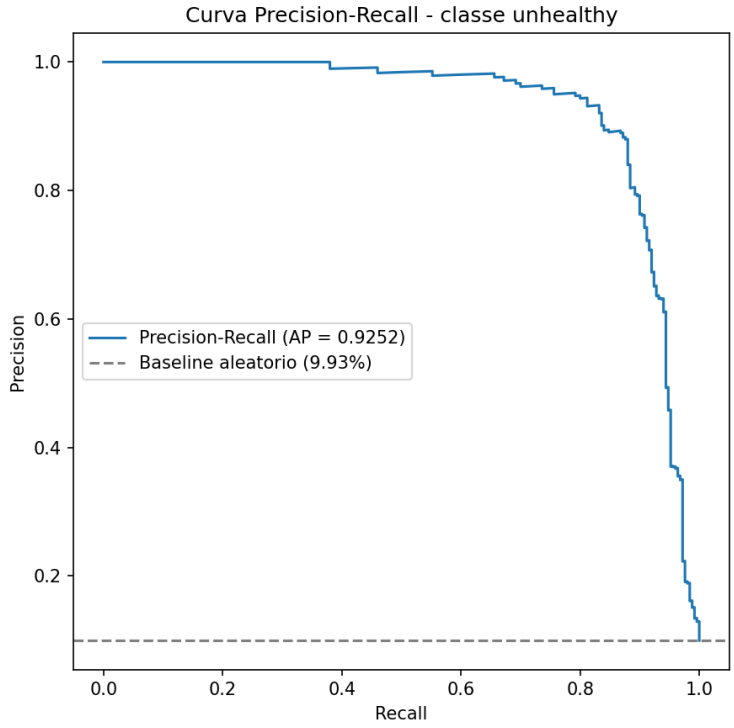


Figura 6.2 - Curva Precision-Recall. Average Precision (AP) = 0.9252.

7. Exemplos de Predicao (split test)



Figura 7.1 - Amostras classificadas pelo modelo: acertos e erros, com classe real e classe prevista indicadas.